

The agent forgot.

Context & memory — why long-horizon agents live or die on state, not prompts. And why a bigger context window won't save you.

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**It captured every
constraint. Then it broke
one nobody removed.**

Week 1, flawless. It logged the rules: *“no prod changes before the 30th.”*

week 3 > proposes a change to the production
tenant on the 22nd

Nobody deleted the constraint. It just **scrolled out
of the context window** and quietly ceased to exist.

“It forgot → give it a bigger window.”

Wrong instinct. Two reasons it backfires:

- **Lost in the middle** — models don't read long context evenly. Bury a rule mid-prompt and you've hidden it, not saved it.
- **Geometric decay** — long horizons rot (success $\approx r^H$). Cramming the context *steepens* the decay.

More tokens is not more memory.

Memory is an **architecture**, not a prompt.

The context window isn't storage. It's **working memory** — scarce, volatile, expensive. Think like an OS:



The agent **pages** information between them — evicting what's cold, retrieving what's suddenly relevant.

Three kinds of memory.

Working

the live window — curated per step, holds only what's needed now

Episodic

what happened — a searchable log of interactions, actions, results

Semantic

durable facts, decisions & constraints — *“prod frozen until the 30th”*

Retrieve, don't **retain**.

Don't accumulate history in the window. Keep it in a store and pull the relevant slice back each step. The hard part is *what* to pull — so score it:

Relevance + Recency + Importance

A hard constraint outranks small talk — and never gets evicted.

Reflect raw history into durable facts.

Week one's hour-long conversation shouldn't stay an hour-long transcript forever. Compress it.

“we talked for an hour” → “**prod is frozen until the 30th**”

Reflection is how episodic memory becomes semantic — how a conversation becomes a **constraint you can always afford to keep.**

A memory design checklist — and the reference architecture.

Named tiers? Working set curated per step?
Constraints extracted to semantic memory?
Retrieval scored? Reflection running? Long
horizon actually tested?

Full post + checklist + architecture diagram — link
in the comments.

Part 4 = accountability. Next: Part 6 — Multi-
Agent: Help vs. Hurt.